EXPERIMENT 4
FLOW PAST A CIRCULAR CYLINDER – VORTEX SHEDDING

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Aim

To simulate two-dimensional flow past a circular cylinder at Reynolds number 100 using the D2Q9 BGK Lattice-Boltzmann Method, visualize the wake and vorticity field, and examine the time variation of drag and lift coefficients and the vortex-shedding frequency.

1. Governing equations and boundary conditions

The supplied solver uses the single-relaxation-time BGK Lattice-Boltzmann formulation on a two-dimensional, nine-velocity (D2Q9) lattice. The particle distribution function for direction i is denoted by f_i.

BGK collision and streaming equation:

$$f_i(x + c_i, t + 1) = f_i(x, t) - (1/\tau)[f_i(x, t) - f_i^eq(x, t)]$$

Macroscopic density and velocity are calculated from the distribution functions:

$$\rho = \Sigma_i f_i$$

$$\rho u = \Sigma_i c_i f_i$$

The second-order equilibrium distribution used by the code is:

$$f_i^eq = w_i\rho[1 + 3(c_i \cdot u) + 4.5(c_i \cdot u)^2 - 1.5(u \cdot u)]$$

Here, c_i are the D2Q9 discrete velocity vectors, w_i are the lattice weights, and τ is the relaxation time. The solver calculates the flow field from these discrete kinetic equations; it does not directly solve a pressure-based Navier–Stokes system

Boundary conditions implemented in the supplied code

Boundary / condition	Implementation
Inlet (left boundary)	Fixed equilibrium distribution with density $\rho = 1$, streamwise velocity $u_x = 0.1$ and transverse velocity $u_y = 0$.
Outlet (right boundary)	Populations at the last column are copied from the adjacent interior column, providing a simple zero-gradient-type outlet approximation.
Cylinder surface	Half-way bounce-back treatment is applied at fluid nodes adjacent to solid cylinder nodes, representing a stationary no-slip wall.
Top and bottom boundaries	The use of array rolling makes these boundaries periodic in the transverse direction. The solver summary identifies this as a periodic far-field approximation.
Initial condition	The flow is initialized with $u_x = 0.1$ and a small sinusoidal transverse-velocity perturbation localized around the cylinder to initiate wake development.

2. Complete parameter set and computational grid

The run used the notebook's LAB mode. The following values are taken from the provided run settings and solver summary; derived values are identified accordingly.

Parameter	Value
Run mode	LAB
Grid size	nx = 180, ny = 80 (180 × 80 lattice nodes)
Reynolds number	Re = 100
Outer time steps	1200
Substeps per outer step	4
Total lattice time steps	4800
Output interval	Every 300 outer steps, plus the final step
Cylinder diameter	10 lattice units
Cylinder radius	5 lattice units
Cylinder centre	x = 45 lattice nodes; y = 39.5 lattice-node coordinate
Inlet lattice velocity	$u_{in} = 0.1$
Kinematic lattice viscosity	$\nu = 0.01$
Relaxation time	$\tau = 0.53$

BGK relaxation rate (derived)	$1/\tau \approx 1.8868$
Final dimensionless time	$Ut/D = 48$
Force calculation	Momentum-exchange method
Strouhal reference	0.16–0.17 for Re = 100

The cylinder is positioned at approximately one-quarter of the domain length. The lattice diameter is calculated in the code as round(ny/8) = 10. The plotted coordinates are scaled by the cylinder diameter: $x/D = i/10$ and $y/D = (j - 39.5)/10$. Thus, the displayed computational region is approximately $0 \leq x/D \leq 17.9$ and $-3.95 \leq y/D \leq 3.95$.

3. Residual or error history

The supplied teaching solver does not calculate or save a conventional iterative residual (for example, a normalized L_2 residual of the governing equations), nor does it compare the computed velocity field with an exact/reference solution at each time step. Therefore, a residual-versus-iteration plot or a formal discretization-error history cannot be reported from the available output. Such a history must not be invented.

Instead, the code records the following time-history diagnostics in cylinder_history.csv at each outer step: drag coefficient Cd, lift coefficient Cl, maximum lattice speed, minimum density, maximum density, and maximum density deviation from unity. The code also checks that the fields remain finite and stops if maximum speed exceeds 0.45 or maximum density deviation exceeds 0.25. These are stability/physicality checks, not residuals and not proof of grid convergence.

Observed diagnostic behaviour in the supplied plots: the drag coefficient has a large initial transient and then settles to an approximately steady level around 1.5–1.7, with small later variations. The lift coefficient initially has small oscillations; larger alternating oscillations become visible at later dimensionless times. The plotted history therefore indicates wake unsteadiness developing during the simulated interval.

4. Required quantitative comparison

The solver reports an estimated Strouhal number St = 0.1666667. The supplied Re = 100 reference interval is 0.16–0.17. The midpoint of this interval is 0.165.

Comparison item	Result
Computed Strouhal number	0.1667
Reference interval	0.1600–0.1700
Reference midpoint	0.1650
Absolute difference from midpoint	0.0017
Percentage difference from midpoint	≈ 1.01%
Assessment	The computed value lies within the stated reference interval.

Percentage difference is calculated as $|0.1666667 - 0.165|/0.165 \times 100$. This is a comparison with the midpoint of the supplied reference range, not a comparison against a single exact experimental value. The estimate is obtained from the dominant frequency in the latter half of the lift history using an FFT. Since the history contains only 1200 recorded outer-step samples and spans $Ut/D = 48$, the estimate has finite frequency resolution.

5. Inline field figure and observed results

Figure 1 shows the final non-dimensional vorticity field generated by the solver. The cylinder is located near $x/D = 4.5$, and the wake extends downstream. The colour scale represents the sign and magnitude of non-dimensional vorticity.

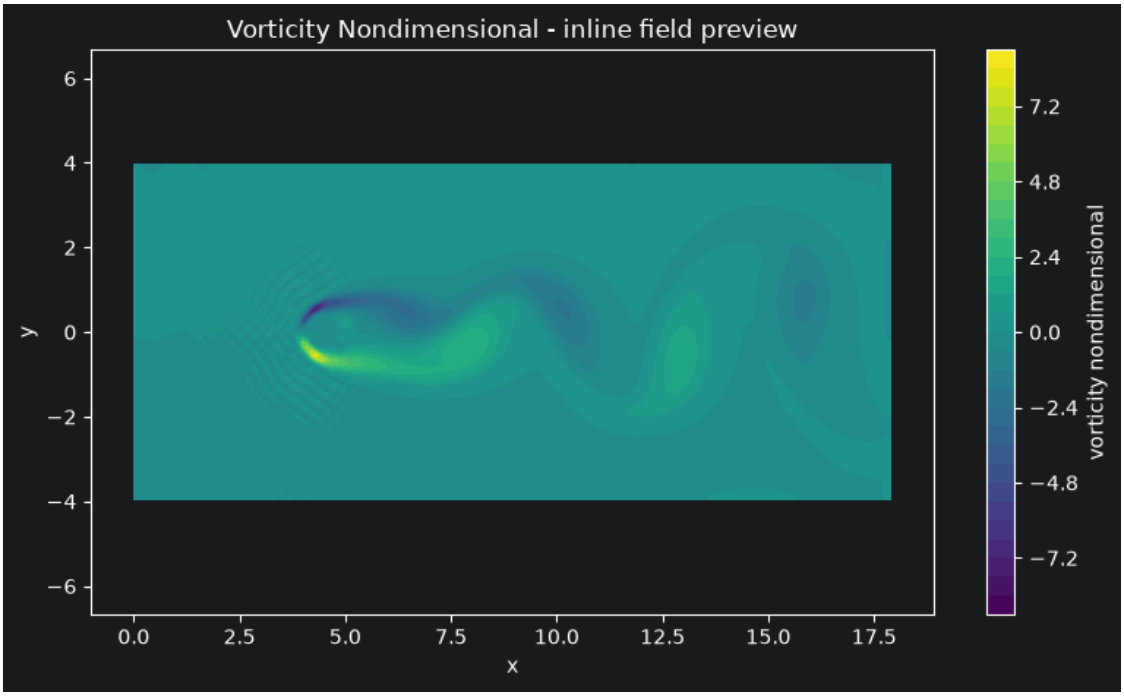


Figure 1. Final non-dimensional vorticity contours for flow past a circular cylinder (Re = 100; 180 × 80 grid).

Observed numerical behaviour: the vorticity plot contains alternating regions of positive and negative vorticity in the downstream wake. The lift curve shows alternating positive and negative excursions, while the drag curve displays a strong initial transient followed by a comparatively flat mean level with smaller fluctuations.

Physical interpretation: the alternating wake structures are consistent with periodic vortex shedding from a bluff body. Alternating circulation produces an oscillating transverse force, which is reflected in the lift coefficient. The streamwise resistance is represented by the drag coefficient; after the start-up transient, its mean value is comparatively stable. These interpretations are consistent with the plotted behaviour, but the plots alone do not establish grid independence or experimental accuracy.

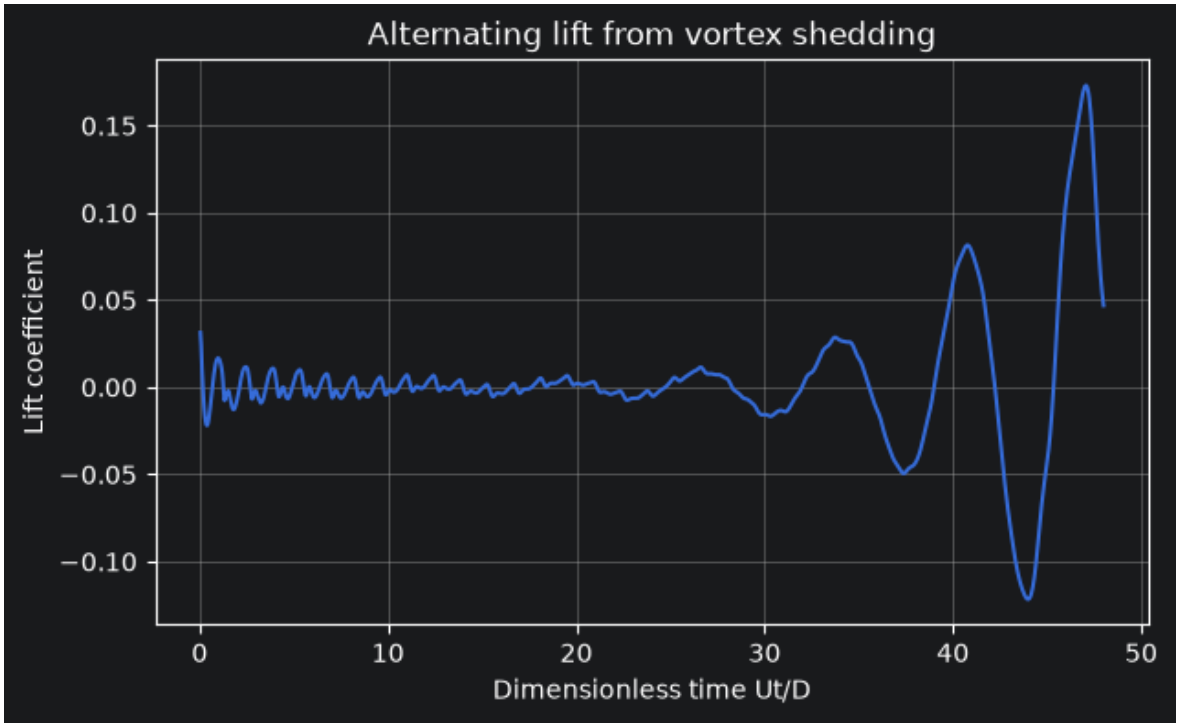


Figure 2. Lift coefficient history versus dimensionless time Ut/D.

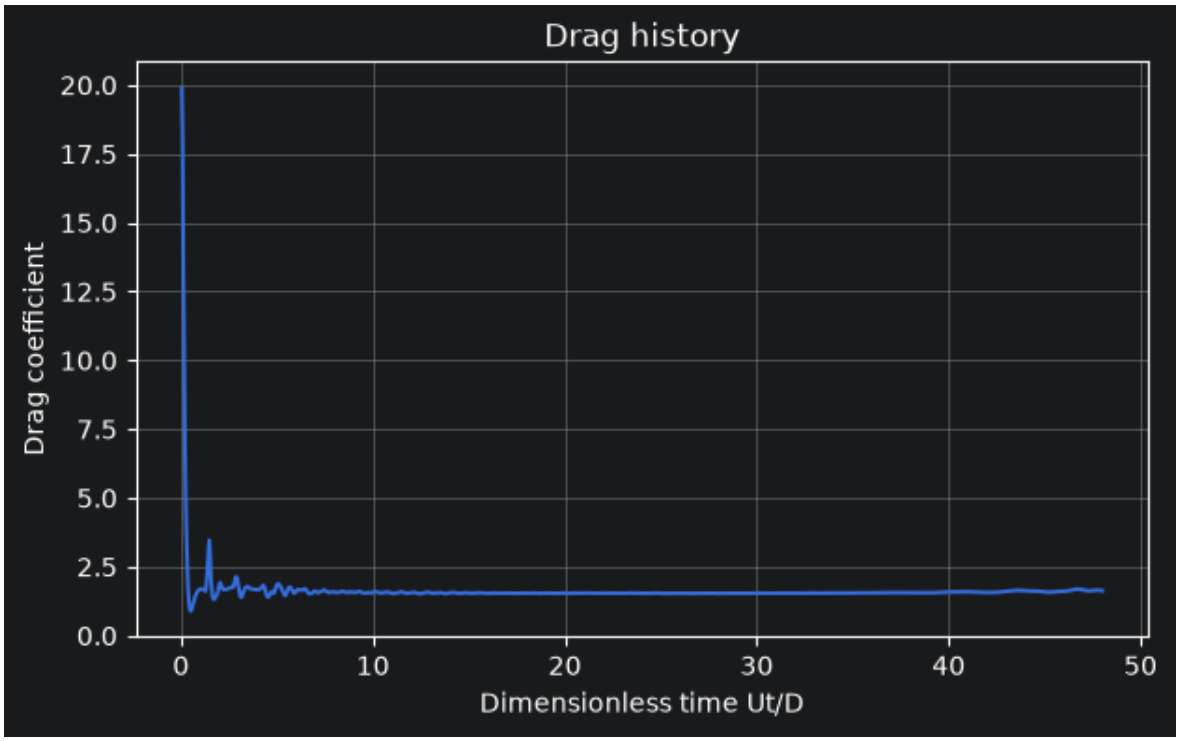


Figure 3. Drag coefficient history versus dimensionless time Ut/D.

6. Limitations of the supplied teaching model

- Coarse spatial resolution: the cylinder diameter is only 10 lattice units. This may not resolve near-wall gradients and the detailed wake as accurately as a finer grid.
- Finite computational domain: the domain is approximately 18D long and 8D high. Boundaries can influence the wake and force coefficients.
- Periodic top and bottom boundaries: these are a far-field approximation, not an unbounded external-flow condition, and may allow wake disturbances to wrap around vertically.
- Simple outlet treatment: copying the adjacent column is a basic zero-gradient-type approximation and may cause reflections or affect downstream flow.
- Low-Mach LBM assumptions: the standard weakly compressible LBM is intended for low lattice speeds. Density variation is monitored, but the supplied report does not provide a detailed compressibility-error study.
- Single-relaxation-time BGK model: one relaxation parameter controls the collision process; more advanced collision models may offer improved stability for some flows.
- Limited simulation duration and frequency resolution: the run reaches $Ut/D = 48$. The FFT estimate uses only the latter half of the lift history, so the frequency estimate is based on a finite record.
- No formal residual or grid-convergence study: the code records force and density/speed diagnostics but does not report equation residuals, truncation error, or a systematic mesh-refinement comparison.
- Simplified force and geometry treatment: forces are evaluated using momentum exchange on the lattice cylinder mask; the discretized circular boundary is not a perfectly smooth continuous cylinder.

Conclusion

The D2Q9 BGK Lattice-Boltzmann simulation was carried out on a 180×80 grid at $Re = 100$ for 4800 lattice time steps. The final vorticity field and lift history show alternating wake structures and unsteady transverse loading, which are characteristic of vortex shedding behind a circular cylinder. The estimated Strouhal number, 0.1667, falls within the supplied reference interval of 0.16–0.17 and differs from its midpoint by approximately 1.01%. However, the model does not provide residual history or a formal grid-convergence assessment, and its coarse grid, finite domain, periodic transverse boundaries, and simplified outlet condition should be considered when interpreting the results.